

Lab 5 - Report

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Group: Mon 11:00-13:00

L3 Switch

1. Create VLANs

```
Multilayer Switch1
Physical Config CLI Attributes

% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
no

Press RETURN to get started!

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name VLAN4
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write memory
Building configuration...
[OK]
Switch#
Switch#
```

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Top

2. Assign ports to VLAN



The screenshot shows a terminal window titled "Multilayer Switch1" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying a series of commands and their outputs. The configuration process involves setting the switchport mode to access and assigning it to VLAN 4. An error occurs when an empty line is entered, followed by a successful configuration of interface fa0/3. The session ends with a "co" command that triggers an "Ambiguous command" warning, followed by a successful "write memory" command.

```
Multilayer Switch1
Physical Config CLI Attributes

Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
      ^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
w
Building configuration...
[OK]
Switch#write memory
Building configuration...
[OK]
Switch#
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
co
% Ambiguous command: "co"
Switch#write memory
Building configuration...
[OK]
Switch#
```

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Top

3. Assign IP Addresses to VLAN Interfaces

Multilayer Switch1

Physical

Config

CLI

Attributes

FastEthernet0/17	unassigned	YES	unset	down
down				
FastEthernet0/18	unassigned	YES	unset	down
down				
FastEthernet0/19	unassigned	YES	unset	down
down				
FastEthernet0/20	unassigned	YES	unset	down
down				
FastEthernet0/21	unassigned	YES	unset	down
down				
FastEthernet0/22	unassigned	YES	unset	down
down				
FastEthernet0/23	unassigned	YES	unset	down
down				
FastEthernet0/24	unassigned	YES	unset	down
down				
GigabitEthernet0/1	unassigned	YES	unset	down
down				
GigabitEthernet0/2	unassigned	YES	unset	down
down				
Vlan1	unassigned	YES	unset	administratively
down down				
Vlan2	2.2.2.1	YES	manual	up
up				
Vlan3	3.3.3.1	YES	manual	up
up				
Vlan4	4.4.4.1	YES	manual	up
up				
Switch#				
Switch#				
Switch#				

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Top

4. Assign IP addresses

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	2.2.2.2
Subnet Mask	255.255.255.0
Default Gateway	2.2.2.1
DNS Server	0.0.0.0

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	3.3.3.3
Subnet Mask	255.255.255.0
Default Gateway	3.3.3.1
DNS Server	0.0.0.0

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	4.4.4.4
Subnet Mask	255.255.255.0
Default Gateway	4.4.4.1
DNS Server	0.0.0.0

IPv6 Configuration	
--------------------	--

5. Enable IP Routing

The screenshot shows a terminal window titled "Multilayer Switch1" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying a table of interface configurations and a series of commands being entered.

Interface	IP Address	Admin Status	Oper Status
FastEthernet0/19	unassigned	YES	unset down
FastEthernet0/20	unassigned	YES	unset down
FastEthernet0/21	unassigned	YES	unset down
FastEthernet0/22	unassigned	YES	unset down
FastEthernet0/23	unassigned	YES	unset down
FastEthernet0/24	unassigned	YES	unset down
GigabitEthernet0/1	unassigned	YES	unset down
GigabitEthernet0/2	unassigned	YES	unset down
Vlan1	unassigned	YES	unset administratively down
Vlan2	2.2.2.1	YES	manual up
Vlan3	3.3.3.1	YES	manual up
Vlan4	4.4.4.1	YES	manual up

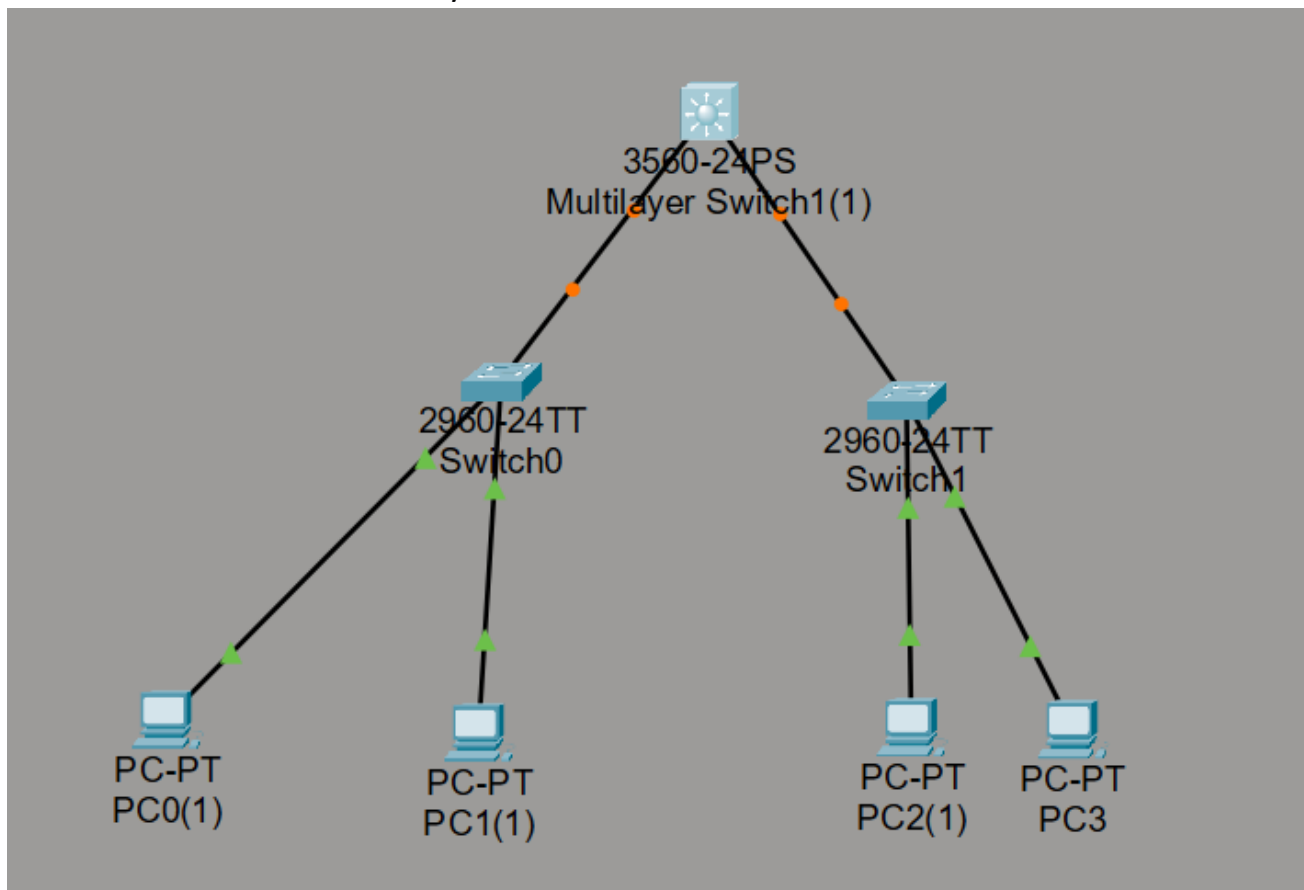
```
Switch#  
Switch#  
Switch#  
Switch#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#ip routing  
Switch(config)#
```

At the bottom right of the terminal window are "Copy" and "Paste" buttons. At the bottom left is a "Top" button.

QUESTIONS:

1. What problem does VLAN segmentation solve in a network?
Separates isolated segments in the network, reduces broadcast traffic and improves security.
2. What does the ip routing command do in a L3 switch?
Enables inter-VLAN routing
3. What is the difference between L2 and L3 Switches?
L2: switches via MAC, no routing.
L3: switches via MAC + routes via IP.

6. L3 Switch as a Distribution Layer



7. L2 Switch 3 and 4

```
Switch#enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#
```

Switch1

Physical Config CLI Attributes

```
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up

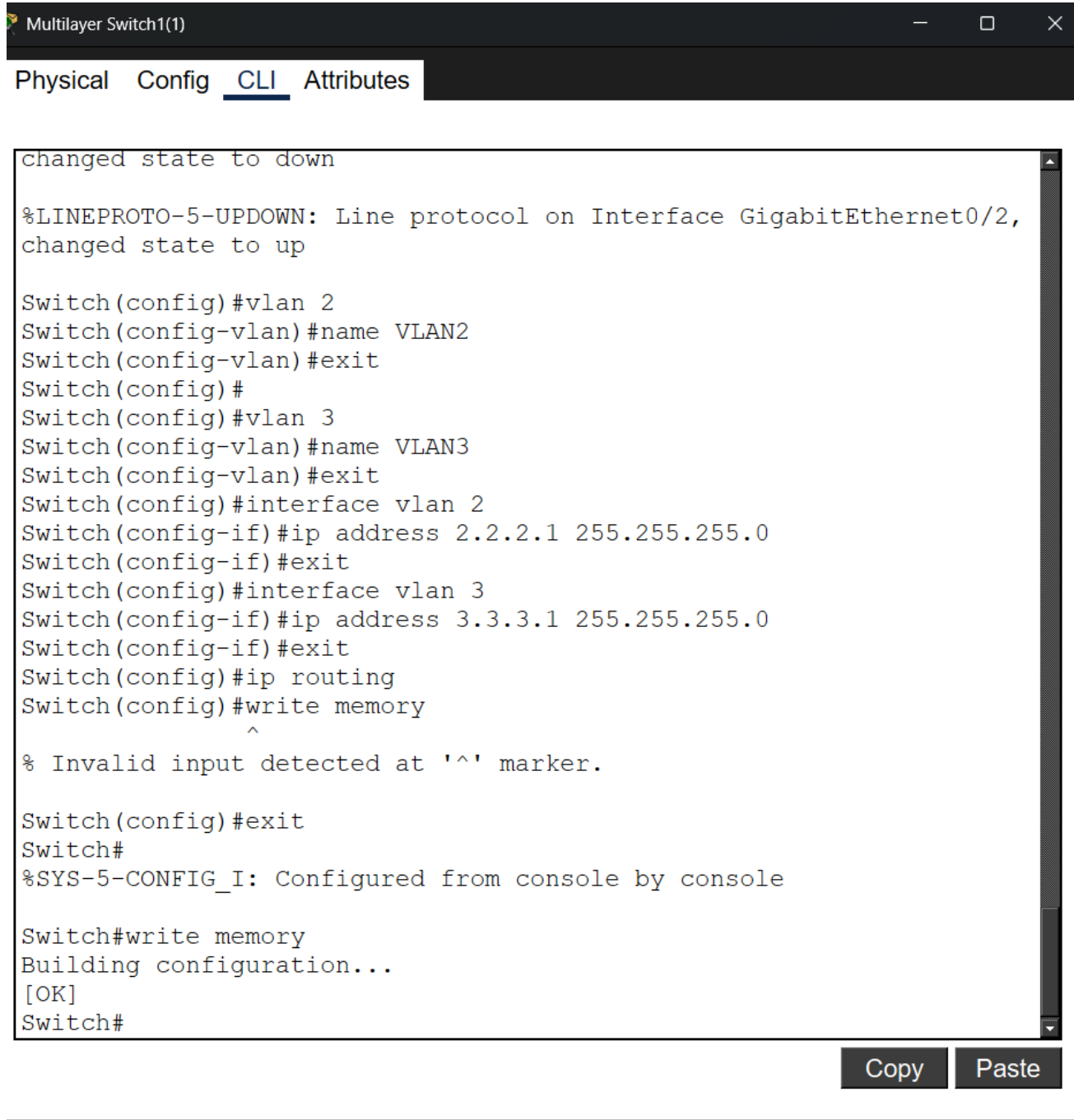
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
% Access VLAN does not exist. Creating vlan 2
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if)#exit
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write memory
Building configuration...
[OK]
Switch#
```

Copy Paste

Top

8. L3 Switch



```
Multilayer Switch1(1)
Physical Config CLI Attributes

changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up

Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#interface vlan 2
Switch(config-if)#ip address 2.2.2.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#interface vlan 3
Switch(config-if)#ip address 3.3.3.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#ip routing
Switch(config)#write memory
^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write memory
Building configuration...
[OK]
Switch#
```

Copy Paste

9. Configuring PCs

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	2.2.2.2
Subnet Mask	255.255.255.0
Default Gateway	2.2.2.1
DNS Server	0.0.0.0

IPv6 Configuration	

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	2.2.2.3
Subnet Mask	255.255.255.0
Default Gateway	2.2.2.1
DNS Server	0.0.0.0

IPv6 Configuration	

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	3.3.3.3
Subnet Mask	255.255.255.0
Default Gateway	3.3.3.1
DNS Server	0.0.0.0

IPv6 Configuration	

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	3.3.3.3
Subnet Mask	255.255.255.0
Default Gateway	3.3.3.1
DNS Server	0.0.0.0

IPv6 Configuration	

10. Test connections

```
Pinging 2.2.2.2 with 32 bytes of data:
```

```
Reply from 2.2.2.2: bytes=32 time=7ms TTL=128
```

```
Reply from 2.2.2.2: bytes=32 time=3ms TTL=128
```

```
Reply from 2.2.2.2: bytes=32 time=4ms TTL=128
```

```
Reply from 2.2.2.2: bytes=32 time=2ms TTL=128
```

```
Ping statistics for 2.2.2.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:
```

```
    Minimum = 2ms, Maximum = 7ms, Average = 4ms
```

```
C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
```

```
C:\>ping 3.3.3.3
```

```
Pinging 3.3.3.3 with 32 bytes of data:
```

```
Reply from 3.3.3.3: bytes=32 time<1ms TTL=128
```

```
Reply from 3.3.3.3: bytes=32 time<1ms TTL=128
```

```
Reply from 3.3.3.3: bytes=32 time=4ms TTL=128
```

```
Reply from 3.3.3.3: bytes=32 time=2ms TTL=128
```

```
Ping statistics for 3.3.3.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

```
C:\>
```

try again.

C:\>ping 2.2.2.3

Pinging 2.2.2.3 with 32 bytes of data:

Reply from 2.2.2.3: bytes=32 time=7ms TTL=128

Reply from 2.2.2.3: bytes=32 time=3ms TTL=128

Reply from 2.2.2.3: bytes=32 time=3ms TTL=128

Reply from 2.2.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 2.2.2.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 3.3.3.2

Pinging 3.3.3.2 with 32 bytes of data:

Reply from 3.3.3.2: bytes=32 time=6ms TTL=128

Reply from 3.3.3.2: bytes=32 time=5ms TTL=128

Reply from 3.3.3.2: bytes=32 time=2ms TTL=128

Reply from 3.3.3.2: bytes=32 time=2ms TTL=128

Ping statistics for 3.3.3.2:

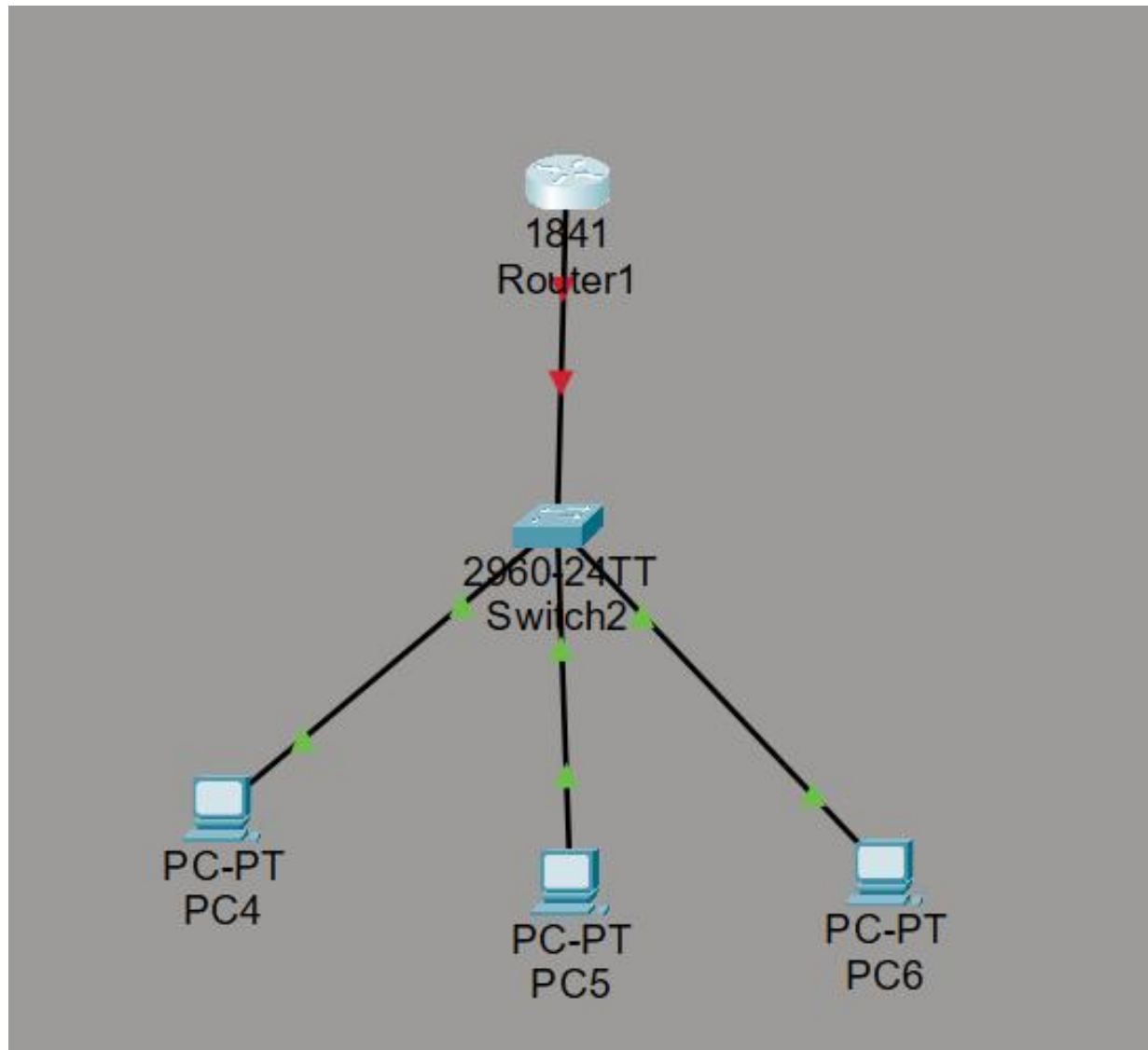
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 6ms, Average = 3ms

C:\>

Part 2: Setting Up a Router for VLAN Communication

11. Switch Configuration



Switch2

Physical Config CLI Attributes

```
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3,4
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write memory
Building configuration...
[OK]
Switch#
```

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Top

12. Router Configuration for Inter-VLAN Routing

```
Router1
Physical Config CLI Attributes

% Invalid input detected at '^' marker.

Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface fa0/0.3
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface FastEthernet0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3,
changed state to up

Router(config)#interface fa0/0.4
Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#end
%LINK-3-UPDOWN: Interface FastEthernet0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.4,
changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
```

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Top

QUESTIONS

1. Why do we configure trunk encapsulation before enabling trunk mode on a Layer 3 switch?

Trunk encapsulation sets the VLAN tagging protocol before trunking can work.

2. Why does the switch automatically create a VLAN when assigning it to a port if it does not exist?

Switch auto-creates VLANs to prevent errors when assigning a non-existent VLAN.

3. If ip routing is not enabled on a Layer 3 switch, how will that impact network communication?

Without IP routing, the switch cannot route between VLANs; inter-VLAN traffic is blocked.

4. What is the difference between an access port and a trunk port on a switch?

Access port carries one VLAN; trunk port carries multiple VLANs.

5. What is the purpose of the encapsulation dot1Q X command on a router?

It tags packets with VLAN ID X for routing across a trunk.

6. Why do we need subinterfaces on a router for inter-VLAN routing?

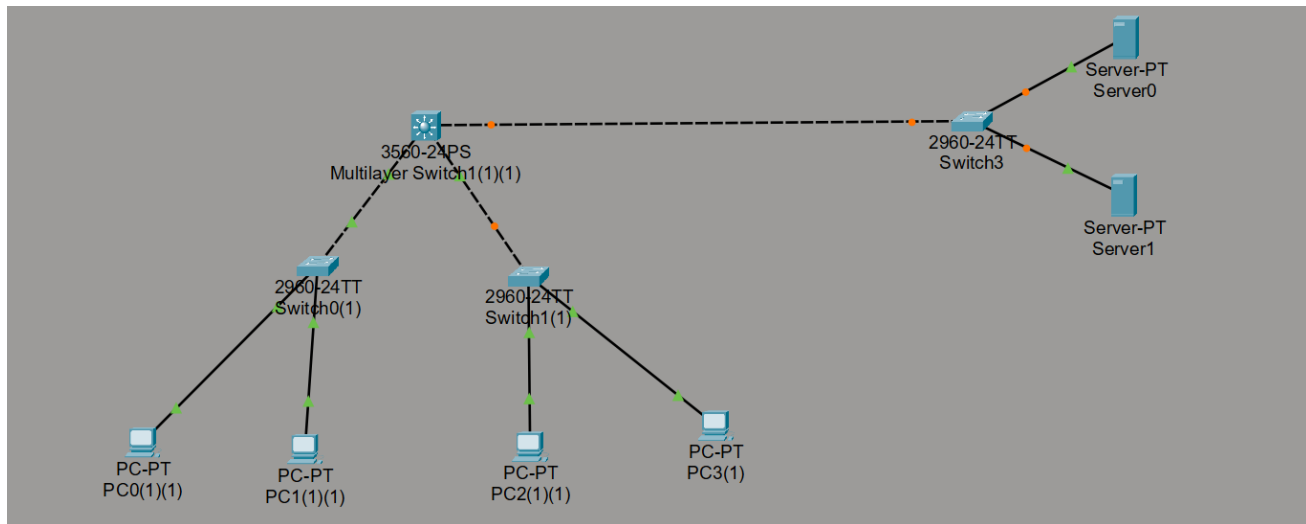
Subinterfaces allow one physical interface to handle multiple VLANs.

7. What is the function of no shutdown command?

Activates an interface; without it, the interface stays administratively down.

TASK:

13. Build



14. Create VLAN 5 on the L3 Switch

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 5
Switch(config-vlan)#exit
Switch(config)#interface vlan 5
Switch(config-if)#ip address 5.5.5.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

15. Configure the Switch Connected to the Servers

```
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 5
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 5
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk allowed vlan add 5
Switch(config-if)#exit
```


16. Configure IP addresses

The image shows two identical screenshots of the 'IP Configuration' window in Cisco Packet Tracer. Each window has a tab for 'IPv6 Configuration' and a section for 'IP Configuration'. In the 'IP Configuration' section, the 'Static' radio button is selected. The fields are filled with the following values:

Field	Value
IPv4 Address	5.5.5.2 (top) / 5.5.5.3 (bottom)
Subnet Mask	255.255.255.0
Default Gateway	5.5.5.1
DNS Server	0.0.0.0

17. Verify Connectivity

```
Command Prompt X
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 5.5.5.2

Pinging 5.5.5.2 with 32 bytes of data:

Reply from 5.5.5.2: bytes=32 time<1ms TTL=128
Reply from 5.5.5.2: bytes=32 time<1ms TTL=128
Reply from 5.5.5.2: bytes=32 time=4ms TTL=128
Reply from 5.5.5.2: bytes=32 time=2ms TTL=128

Ping statistics for 5.5.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 5.5.5.3

Pinging 5.5.5.3 with 32 bytes of data:

Reply from 5.5.5.3: bytes=32 time<1ms TTL=128
Reply from 5.5.5.3: bytes=32 time<1ms TTL=128
Reply from 5.5.5.3: bytes=32 time<1ms TTL=128
Reply from 5.5.5.3: bytes=32 time<1ms TTL=128

Ping statistics for 5.5.5.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```